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PALEOMAGNETISM OF THE SAMAIL OPHIOLITE, OMAN
2. THE WADI KADIR GABBRO SECTION

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Abstract. Four stratigraphic (vertical) transects were obtained from the gabbroic and ultramafic section of the Samail ophiolite (middle Cretaceous age). The transect through Wadi Kadir was the most detailed; 65 independent oriented cores were obtained from 22 sites on a 14-km traverse. From field mapping we infer that this traverse corresponds to a vertical section of 6.0 km, which includes sheeted dikes, plagiogranites, isotropic high-level gabbro, transitional gabbro, cumulate gabbros, cumulate ultramafics, and ultramafic rocks. There are some noticeable variations in magnetic mineralogy with depth: (1) primary exsolved magnetite is found mainly in the late stage high-level gabbro, and (2) secondary magnetite is found mainly in the upper 3.5 km of the cumulate gabbros and in the cumulate ultramafics. Most of the gabbros have a stable magnetization direction that is almost horizontal and slightly west of north. The median natural remanent magnetization (NRM) intensity for the cumulate gabbros is 0.93 A/m. The median Q_n value is 3.0 with most values falling in the range of 1 to 5. The ultramafic rocks below the gabbroic layer are more strongly magnetized due to higher concentrations of secondary magnetite, and their magnetization is more stable to AF demagnetization than that of the overlying gabbros. It is not clear what fraction of their magnetization was acquired beneath the seafloor because some of the serpentinization is thought to be of subaerial origin. Hysteresis measurements on the gabbros suggest that the magnetic carriers in the gabbros are a few tenths of microns in diameter (pseudo single domain). The magnetic properties of the cumulate gabbro section divide it into an upper intensely magnetized layer (3.5 km thick) and a lower weak layer (2 km thick). This was also found for the Point Sal ophiolite (California). The low intensities below 2 km are in agreement with petrological observations of a minimum in the concentration of secondary magnetite. The upper intense layer can contribute over one half of the amplitude of any marine magnetic anomaly which might have existed when the ophiolite was submarine. The serpentinized rocks also are possible contributors to anomaly amplitude. Consideration of the probable variations in the magmatic and hydrothermal history of the ocean crust and comparison of our results to other studies indicate that it may not be possible to specify a generalized magnetic source model for the plutonic section of the ocean crust.

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Introduction

In this paper we will investigate the possibility that oceanic layer 3 gabbros are capable of providing a significant contribution to measured marine magnetic anomalies. In doing so, we will put aside the controversies surrounding the origin of ophiolites [Miyashiro, 1975; Moores, 1975; Pearce et al., 1981] and assume a priori that the Samail ophiolite was formed at a mid-ocean spreading center. Our magnetic measurements cannot provide data to help resolve the ophiolite controversy, but they can yield useful information concerning the magnetic structure of the deeper layers of the oceanic crust. Two classes of information are available on the magnetic structure of the oceanic crust. First, there are methods related to the interpretation of magnetic anomalies using either forward modeling [Vine and Matthews, 1963; Talwani et al., 1971; Blakely, 1976; Cande and Kent, 1976] or inversion techniques [Parker and Huestis, 1974]. Although these methods have met with some success, they are hampered by the nonuniqueness problems associated with any potential field calculation. Second, there are the direct measurements of the magnetic properties of ocean crust samples. Direct measurements have been made on dredge haul samples, Deep Sea Drilling Project (DSDP) drill cores, and more recently, samples returned from Alvin and Cyana submersible dives. A diverse suite of rocks have been recovered, but the collections are biased toward samples from the uppermost layers. Each collection technique has its own drawbacks that make the task of representative sampling extremely difficult. DSDP cores have provided the most detailed information to date, but drilling technology limits the recovery to samples in the upper section of the crust. Nevertheless, Dunlop and Prévot [1982] were able to assemble a collection of intrusive rocks from DSDP legs 30, 37, and 45 in the North Atlantic for magnetic study. Limited bottom sampling of zero-age crust has been achieved during recent submersible dives on the Mid-Atlantic Ridge [Johnson and Atwater, 1977], and some of these samples have been oriented. Dredging is not a preferred method of sampling but is the only way at the present time of obtaining samples from the deeper intrusive layers. Fox and Opdyke [1973] and Kent et al. [1978] have investigated the magnetic properties of gabbros dredged from North Atlantic fracture zones. Their information is useful, but the results are difficult to interpret for the following reasons:

1. The association of the samples with fracture zones may have altered the magnetic properties.
2. There is no stratigraphic control.
3. The samples are not oriented.

Our study of the Samail gabbros overcomes the limitations of reasons 2 and 3 above, but we are

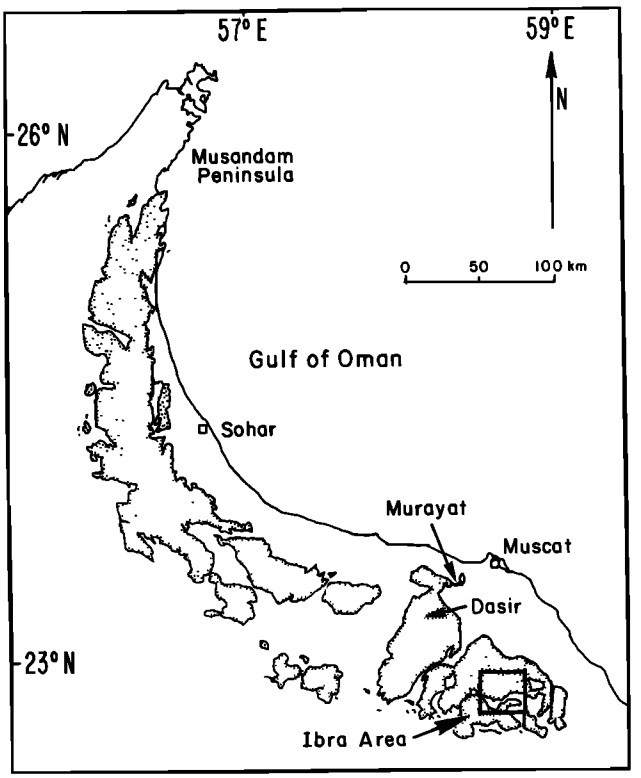


Fig. 1. Map of the Sultanate of Oman region showing the Oman Mountains, Samail ophiolite, and the study area near Ibra.

faced with the problems of a similar nature to reason 1; i.e., are the magnetic properties we measure representative of oceanic layer 3? It is not difficult to imagine alteration of the magnetic state during obduction and subsequent structural reorganization of the ophiolite. The ultimate goal of our work is to document the magnetic properties of layer 3; in this paper we present the results that we have obtained on our way to this goal. Our approach consists of two parts. First, we must document the paleomagnetic properties of the gabbros and show that (1) the intensity of natural remanent magnetization (NRM) is sufficient to give a significant contribution to the observed anomaly, and (2) the directions of magnetization are coherent throughout most of the gabbroic layer. Second, we must use rock magnetism to isolate the nature of the remanence carriers and show that the intensities and directions that we have measured are indeed representative of preobducted layer 3 rocks.

A thorough reconnaissance study of the Samail ophiolite was reported by Glennie et al. [1974] and several detailed descriptions of the ophiolite (and our field area) can be found in the Journal of Geophysical Research Oman ophiolite volume [Hopson et al., 1981]. Radiometric ages for the section near our study area are between 94 and 97 m.y. [Tilton et al., 1981]. The ophiolite occurs as a northwest trending massif along the Gulf of Oman (Figure 1). In the study area (Figure 2) the gabbros are exposed in a series of wadis or canyons that cut north-south through the ophiolite. Four transects were made through the gabbro section (Figure 2). In this paper we will present only the results from Wadi Kadir. The sample locations for this transect are labeled L5 through L26 (Figure 2). Sixty-five independent core samples were obtained from these sites covering a 14-km traverse. From the field data obtained by Hopson et al. [1981] we infer that our traverse corresponds to a vertical section of 6.0 km (Figure 3). The gabbros were much easier to drill than the dike sequence, and at most sites it was possible to recover at least a 5-cm core sample, providing two subsamples 25 mm in length and 25 mm in diameter. Each core sample was oriented using a compass and orienting tube. All data reported here are for independent core samples.

Pallister and Hopson [1981] studied four stratigraphic sections through the ophiolite. One of these sections is the Wadi Kadir II section, which they helped us sample. They recognized eight different rock units in their sections, all of which are represented in the Wadi Kadir II section (Figure 3). At the section top are sheeted dikes altered to greenschist facies metamorphic rocks [see Pallister, 1981; Luyendyk et al., this issue]. Beneath this unit are three rock units representative of the top of the oceanic layer 3 magma chamber. These units include noncumulus high-level gabbro which has undergone hydrothermal alteration, a transitional gabbro which has cumulus textures but is not layered, and plagiogranites which are late-stage differentiates of the high-level gabbro. These three units are less than 1 km thick at Wadi Kadir; they are believed to postdate the sheeted dike sequence. Hydrous alteration decreases downward through these units. Beneath them are 5.0 to 5.7 km of cumulate layered gabbro. The primary cumulus phases are olivine, clinopyroxene, and plagioclase. This unit is comparatively little-altered relative to the overlying units. No intrusive contacts were seen in the cumulate layered gabbro, suggesting crystallization from a single large long-lived magma chamber which crystallized from the base upward. Below the cumulate gabbro are found two thin (0.2 km) cumulate ultramafic units; a cumulus wehrlite above cumulus dunite. These are the first rock units crystallized from the magma chamber. They overlie tectonitic harzburgite and dunite mantle rocks. The base of the measured section is taken at the base of the dunite and is interpreted to be the base of the crustal magma chamber.

Pallister and Hopson [1981] completed two measured sections in Wadi Kadir; a western (I) and an eastern (II) section. Locations of these sections are shown in Figure 2 of Pallister and Knight [1981]. Pallister and Hopson [1981] show the results of their study of the western section (I), whereas our samples are from the eastern section (Wadi Kadir II). The sections differ only in the thickness of rock units above a fault at the 3.4 to 4.0 km stratigraphic level (Figure 3). Above this fault the western section measures 700 m thicker in the cumulate gabbro compared to the Wadi Kadir II section [see Pallister and Hopson, 1981, Figures 5 and 6].

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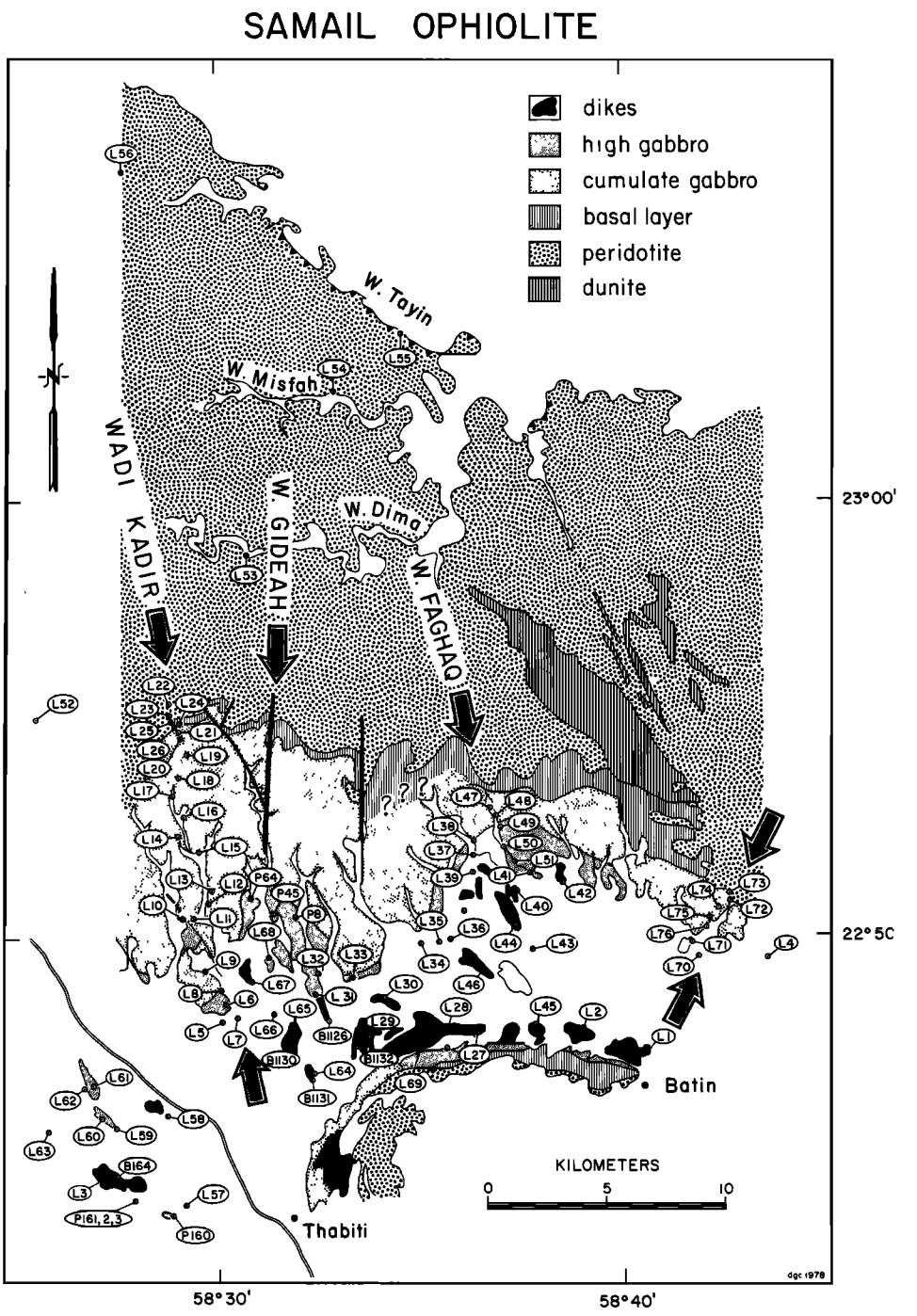


Fig. 2. Simplified geologic map of the ophiolite nappe near Ibra and Thabiti. Magnetic sampling sites L1 through L76 are shown. Broad arrows indicate traverses through the gabbro section.

Experimental Results

Intensity of NRM, Susceptibility, and Koenigsberger Ratio

The NRM intensity in the cumulate gabbros varies from 3 x 10^{-2} A/m to 6 A/m but most of the samples have an NRM intensity close to the median value of 0.93 A/m (mode = 1.09 A/m). The NRM intensity for 61 cores as a function of stratigraphic level is shown in Figure 3. The NRM intensity at site L13 is extremely high (60, 106, 329, and 969 A/m), and these data are omitted from Figure 3. Site L13 samples also have high susceptibility and J_S values which suggest an unusually high concentration of magnetic material in the samples. Although no systematic variation of NRM with depth is apparent, there is a

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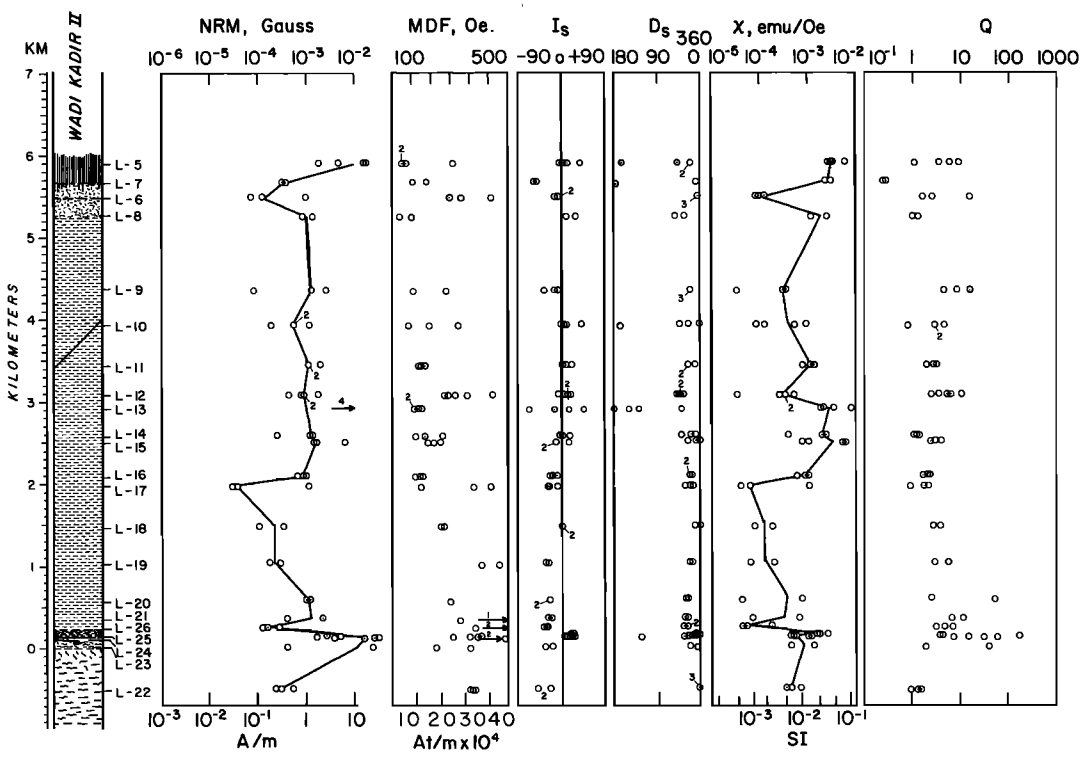


Fig. 3. A reconstruction of the vertical stratigraphic section (eastern or II) at Wadi Kadir. The paleomagnetic results are plotted as a function of stratigraphic position in the section. Both cgs and SI units are given. Level zero is the hypothesized crust-mantle boundary. From the top of the section downward, the lithologic symbols indicate (1) sheeted dikes, (2) noncumulus high-level gabbro (L7 is plagiogranite), (3) transitional nonlayered cumulus gabbro, (4) layered cumulus gabbro, (5) wehrlite-melagabbro cumulates, (6) dunites, (crust-mantle interface), and (7) harzburgite plus dunite. Stratigraphy courtesy of J. Pallister. Pallister and Hopson [1981] describe the petrography and chemistry of part of this section. The vertical trends for NRM and susceptibility are drawn through the log of the median value at each site. NRM and Q are not plotted for site L13, which may contain an unusually high concentration of magnetic minerals.

suggestion of about a factor of 3 decrease in NRM intensity (and susceptibility) toward the bottom of the gabbro section (0-2 km below L16; Figure 3), which divides it into two magnetic layers [Day and Luyendyk, 1979]. This can be explained by petrographic observations of a decrease in the concentration of opaque minerals versus depth [Pallister and Hopson, 1981; J. S. Pallister, personal communication, 1981].

The geometric mean (The geometric mean is the antilog of the average of the log (base 10) of the individual values. Values of the geometric mean are very close to the median value for our sample set.) intensity for the Wadi Kadir gabbro samples (1.0 A/m; Table 1) is higher than the geometric mean values found for dredge gabbros (0.3 A/m; Kent et al. [1978]) and close to the values for the Troodos complex (1.0 A/m arithmetic mean; Vine and Moores [1972]), but is 10 or more times higher than the mean NRM intensity for gabbros from the Point Sal [Kempner and Luyendyk, 1976] (Table 2), Vourinos, Macquarie Island, Smartsville [Banerjee, 1980], and Othris ophiolites [Beske-Diehl and Banerjee, 1979]. For the Bay of Islands ophiolite, Swift and Johnson [1982] have also divided the gabbro section into two magnetic layers. They found values near 3 A/m in the upper section and near 0.02 A/m in the lower. Dunlop and Prevot [1982] proposed a magnetization model, based on DSDP samples, where upper amphibolite-grade gabbros have a NRM of 1.0 A/m and lower fresh gabbros have 0.2 A/m. As we show in Table 2, our Samail gabbro results give a median value of 0.9 A/m for upper gabbros (above 2 km) and 0.27 A/m for lower gabbros. In our discussion below we propose that this magnetic stratification is a natural result of the magmatic history and retrograde metamorphism of the crust near the spreading center.

The NRM value is quite high for the wehrlites and dunites. This is due to high concentrations of secondary magnetite in these rocks from the serpentinization of olivine.

The weak field susceptibility χ (Figure 3) and the NRM intensities were used to calculate the Koenigsberger ratio Q_n, $Q_n = NRM/\chi H$, where H is the intensity of the earth's magnetic field taken as 0.5 Oe, or 3.97×10^4 A turn/m. Q_n is a measure of the relative importance of the remanent magnetization in relation to the induced magnetization (χH) and a value greater than 1 is indicative of a remanence-dominated magnetization. The Q_n values in the section are dominated by values greater than 1; they range from 0.87 to 348. The geometric mean value in the gabbros is 4.86 (Table 1), and is comparable to the values

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TABLE 1. Statistics of Some Magnetic Properties of Plutonic Rocks from the Wadi Kadir Section

	Plagioclase Granites	High-Level Gabbros	Transitional and Cumulate Gabbros (Upper and Lower)	Layered Ultramafics	Dunite	Peridotite
			NRM, A/m			
Mean	0.35(2)	0.39(3)	34.9(44)*	3.82(2)	21.83(3)	5.47(7)
σ		0.50	0.16		5.10	11.5
Median		0.12	0.93		23.3	0.51
Geometric mean		0.20	1.00		21.4	1.14
			MDF, Oe			
Mean	178(2)	390(3)	273(42)	355(2)	428(3)	489(7)
σ		108	181		25.6	181
Median		360	232		435	425
Geometric mean		380	223		428	459
			Susceptibility+, SI			
Mean	4.5 E-2(3)	1.9 E-2(4)	1.4 E-2(45)	2.3 E-2(3)	1.4 E-2(3)	5.9 E-3(7)
σ	2.0 E-2	3.4 E-2	2.0 E-2	6.5 E-3	1.7 E-3	1.2 E-3
Median	3.9 E-2	1.4 E-3	6.7 E-3	2.0 E-2	1.4 E-2	5.6 E-3
Geometric mean	4.3 E-2	3.5 E-3	5.5 E-3	2.3 E-2	1.4 E-2	5.8 E-3
			Q_n			
Mean	0.26(2)	6.33(3)	21.9(44)*	3.86(2)	40.0(3)	24.0(7)
σ		7.38	64.6		12.6	51.7
Median		2.55	3.0		37.0	1.80
Geometric mean		3.94	4.86		38.7	4.90
			H_C, Oe			
Mean	141(1)	96(1)	152(16)	211(1)	141(1)	211(2)
σ			43.5			
Median			159			
Geometric mean			144			
			H_{RC}, Oe			
Mean	330(1)	285(3)	275(16)	300(1)	225(1)	293(2)
σ		10.0	69.6			
Median		275	270			
Geometric mean		285	265			
			J_R/J_S			
Mean	0.11(1)	0.28(1)	0.21(16)	0.33(1)	0.23(1)	0.28(2)
σ			6.4 E-2			
Median			0.21			
Geometric mean			0.20			

Values in parentheses are the number of samples.
*Skewed positive due to high values at L13.
+A few additional values are from other nearby locations.

for dredged gabbros (Q = 3; Kent et al. [1978]), drilled gabbros (Q = 2.5-12.7; Dunlop and Prévot [1982]), the Troodos ophiolite (Q = ~5-10; Vine and Moores [1972]), and the Bay of Islands ophiolite (Q = 1.6-3.6; Swift and Johnson [1982]), but higher than the values at other ophiolites where the Q_n value is generally less than 1 [Kempner and Luyendyk, 1976; Beske-Diehl and Banerjee, 1979; Banerjee, 1980]. Significantly, the Q_n for gabbros is higher than the value of 1 found for the sheeted dikes here [Luyendyk et al., this issue].

Notice that although NRM and χ are lower in the basal gabbros, the Q_n ratio is about the same here as in the higher gabbros (Figure 3). Therefore NRM and χ are varying proportionally and probably reflect the concentration of magnetic minerals.

Opaque Mineralogy and Thermomagnetic Curves

Pallister and Hopson [1981] performed petrographic and microprobe studies of Fe-Ti oxides. The detection of these oxides in thin section was largely restricted to stratigraphic levels above 2 km [Pallister and Hopson, 1981, Figure 6].

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TABLE 2. Total Magnetizations (Median Values)

	NRM J, A/m	$Q_n(0.5)$	~Thickness Δt, m	Total Magnetization, A	Anomaly Relief for 5-km Ocean**, nT
Extrusive basalts * (n=22)					
Samail	1.51	>1.6	1000	1510	110
Point Sal+	0.4	0.72	1300	520	
Sheeted dikes (n=202)					
Samail	0.73	0.8	1000	730	45
Point Sal	0.34	0.26	100	34	
Plagiogranite plus high-level, transitional, and upper cumulate gabbros (n=35)					
Samail	1.03	2.84	3700	3811	174
Point Sal	0.12	0.33	>500	>60	
Lower cumulate gabbros (n=14)					
Samail	0.27	3.32	2000	540	18
Point Sal	0.02	0.47	>550	>11	
Total					347

*Our work on 22 samples.

+Point Sal values are geometric means.

**The magnetization is doubled in this calculation to simulate the contrast of adjacent oppositely magnetized crust.

Both primary oxides and those resulting from the serpentinization of olivine (secondary) were seen. Microprobe studies show that the primary oxides are magnetite and ilmenite and possibly ilmenohematite and that the secondary oxides are pure magnetite [see Pallister and Hopson, 1981, Table 7]. The primary oxides are mostly restricted to the high-level gabbros and are exsolved titanomagnetites.

Our own reflected light studies on a different set of samples confirm the findings of Pallister and Hopson [1981]. In addition, we found common alteration of magnetite to hematite in the high-level gabbros. Sulfides were also detected in the high-level gabbros.

We determined the percentages of primary and secondary oxides from thin sections of our samples (Figure 4). Primary oxides are most abundant in the high-level and transitional gabbros above 5 km; secondary magnetite correlates in occurrence with olivine [Pallister and Hopson, 1981, Figure 6] and is detected largely in the upper two thirds of the layered gabbro, above 2 km. Given that olivine is largely uniformly distributed through the cumulate gabbros, it appears that the alteration of it to magnetite is more pronounced in the upper section (L9-L16, Figure 3). Gregory and Taylor [1981] have demonstrated that seawater has thoroughly penetrated the ophiolite section here during hydrothermal circulation (T>400°-500°C) at the rise crest. This process could have produced secondary magnetite from Fe, Mg silicates. However, these silicates are more Fe-rich in the upper cumulates, which results in more magnetite produced here during serpentinization (see discussion section). Similar NRM and opaque mineral concentration variations were observed in the gabbro section at Point Sal, California [Kempner, 1977].

Thermomagnetic curves were determined for 11 gabbro and ultramafic samples (Figure 5). The curves were run on magnetic separates at 5×10^3 Oe (3.98×10^5 A turn/m) and a vacuum of 10^{-2} torr. The curves are reversible and give Curie temperatures from 548°C to 578°C. Some Ti content in the magnetite is suggested. The microprobe results of Pallister and Hopson [1981] suggest that this may be about 5%. Two notable attributes are seen in the thermomagnetic curves of L17.01 and L25.03. The curve for L17.01 (cumulate gabbro) and for a sister sample show a concave shape, suggesting paramagnetic material. This sample lies at the 2-km limit of detection for secondary magnetite (Figure 4). The curve for L25.03 (cumulate ultramafic) shows a slight hump in the heating curve between 200°C and 350°C. This may indicate the presence of sulfides, as noted by Dunlop and Prévot [1982] for a serpentinized gabbro from DSDP leg 37.

Hysteresis Experiments

Hysteresis measurements on 22 selected samples (Figure 6) indicate that the magnetic carriers are in the pseudo-single domain (PSD) size range with coarser varieties in the high gabbros and finer in the lower cumulates. This can be seen in the plots of the ratios of remanent coercive force H_{RC} to coercive force H_C and saturation remanence J_R to saturation magnetization J_S (Figure 7). The shapes of the hysteresis curves vary with stratigraphic level. Above 2.0 km from the harzburgite contact (L16) the high field section of the curves flatten or saturate (e.g., L8.01A and L13.02 in Figure 6); below this level the high field portions show a positive slope, suggesting relatively more paramagnetic material (e.g., L19.01A and L22.01A). In addition, H_{RC} and J_R/J_S are slightly higher below 2.0 km than

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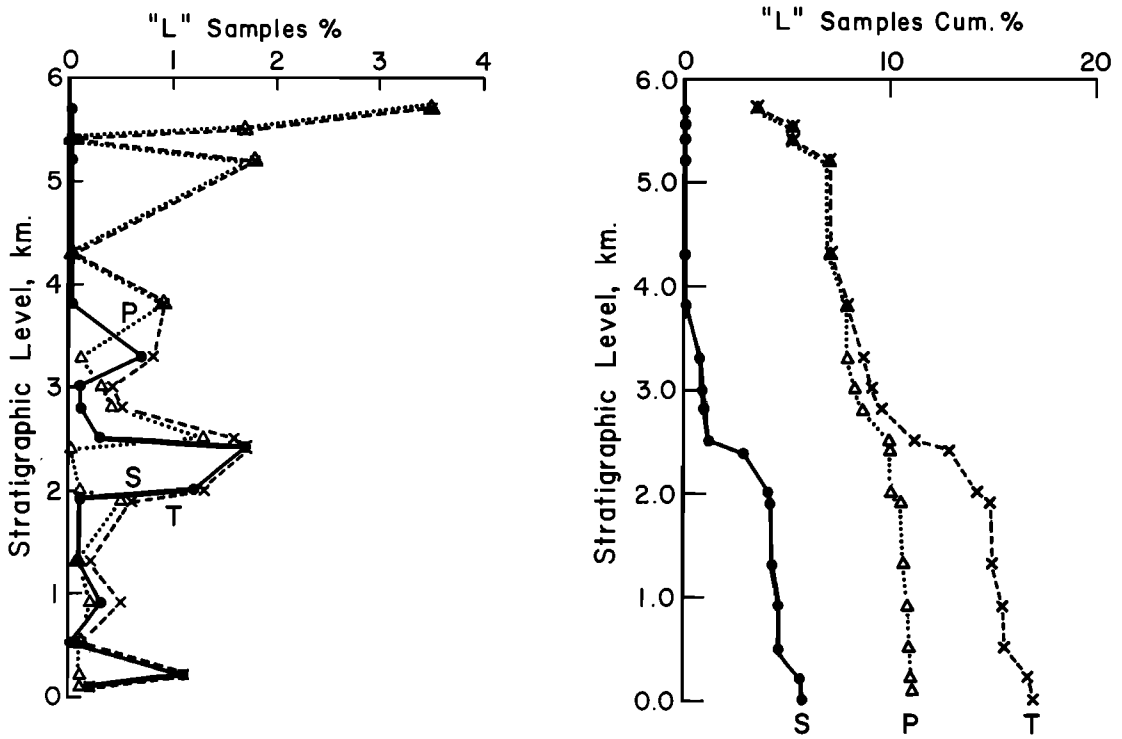


Fig. 4. The concentration of opaque minerals observed in thin sections of the magnetic samples, plotted against stratigraphic level. On the left is the concentration versus level and on the right is the cumulative concentration summed from the top down. The total percentage of opaques is 'T' (dashed line), the primary percentage is 'P' (dotted line), and the secondary percentage is 'S' (solid line).

above this level (Figure 8), suggesting a somewhat finer grain size for magnetic carriers in the lowermost cumulate gabbros. The stratigraphic level of these contrasts correlates with the level of the decrease in NRM intensity and susceptibility (Figures 3 and 8) plus the decrease of secondary magnetite observed in thin section. The two classes of hysteresis curves found here have also been found for dredged ocean gabbros and in the same stratigraphic order in the gabbros from the Point Sal ophiolite [R. Day, unpublished data, 1980].

The J_R/J_S-H_{RC}/H_C plot for the gabbros and ultramafics (Figure 7) shows that these ratios are not well inverse correlated, contrary to the results of Day et al. [1977] for synthetic-sized titanomagnetites and our results for the Samail sheeted dike sequence [Luyendyk et al., this issue]. Dunlop and Prévot [1982] also noticed a lack of systematic inverse correlation among their drilled samples. Anomalous samples in Figure 7 are those with H_{RC}/H_C values near 3, while at the same time demonstrating high J_R/J_S values near 0.3. These anomalous samples are from the upper high-level gabbros and lower cumulate gabbros but not from the upper cumulate gabbros. It is not presently clear how to interpret correctly these results.

Stability of NRM to Demagnetization

The gabbros show a wide range of stability to alternating field demagnetization (AFDM). The median demagnetizing fields (MDF) are shown versus stratigraphic level in Figure 3, and representative demagnetization curves are given in Figure 9. The magnetization directions of the samples are also shown at sequential demagnetization values (Figure 9). The ultramafic rocks are more resistant to AFDM than the gabbroic rocks primarily due to the finer grain size of magnetic carriers they contain (Figures 7 and 8). Both the gabbroic and ultramafic rocks show multicomponent magnetization directions (Figure 10), but the ultramafic rocks and some lower cumulate gabbros may have a hard secondary component which cannot be removed or have acquired a anhysteretic remanence during AFDM. A stable magnetic direction could be obtained from the majority of gabbroic rocks.

Thermal demagnetization (TDM) was completed on six gabbroic rocks and three ultramafic rocks (Figure 11). Both groups of samples show maximum blocking temperatures from 550°C to 580°C. Most of the samples show a narrow range of blocking temperatures. In most cases the stable magnetic direction selected during TDM was very close to the stable direction selected during AFDM of sister samples (Figure 11). A dramatic exception to this is peridotite sample L24.02.

The TDM experiments on the plutonic rocks contrast very sharply with TDM experiments on the overlying sheeted dike sequence [Luyendyk et al., this issue]. The TDM of dike samples showed that about 30% of the remanence in some samples is carried by hematite. This remanence is of reversed polarity compared to normal polarity carried by magnetite in the dikes. The plutonic samples tested here do not appear to have remanence in any high blocking temperature carriers,

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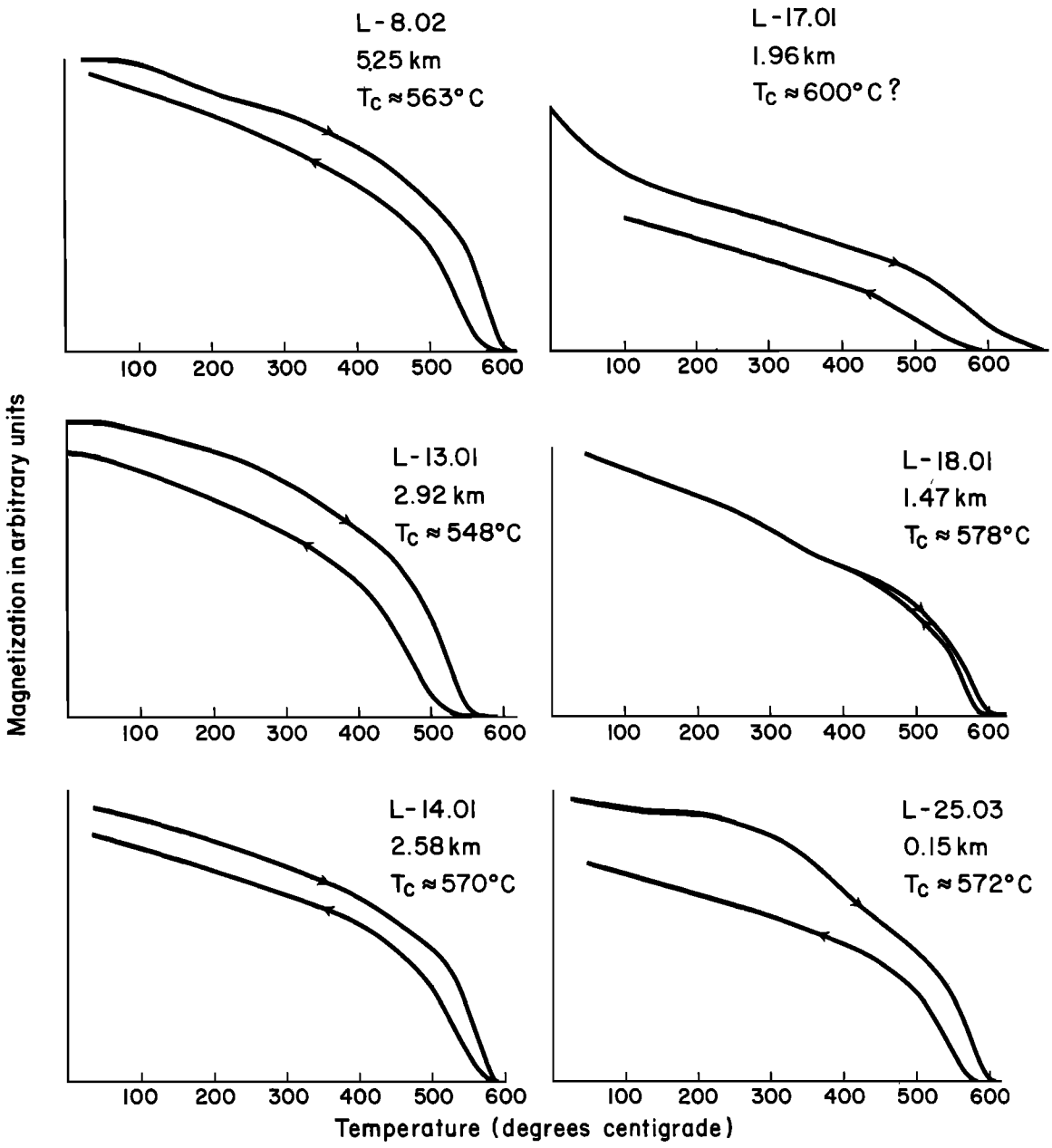


Fig. 5. Representative thermomagnetic curves run on the Wadi Kadir samples. L8 is a transitional gabbro, L13 and L14 are upper cumulate gabbros, L17 and L18 are lower cumulate gabbros, and L25 is a cumulate ultramafic rock.

i.e., hematite. The magnetization direction in the plutonic section is normal (see below) and is evidently carried in primary and secondary magnetite. The observation of R magnetization carried by hematite in the hydrothermally altered dikes suggests that the hematite is of submarine rather than subaerial genesis. Presumably, if hematization occurred at the outcrop, it would have occurred in both the hypabyssal (dikes) and plutonic (gabbro) section.

Paleomagnetic Directions

The stable paleomagnetic directions for 49 gabbro samples from sites L7 down to L26 and 12 ultramafic samples from L22 to L25 (I_S and D_S, Figure 3) are shown on stereograms in Figure 12. The stable directions are selected by joint inspection of the normalized demagnetization curves, the stereograms of directional change, and orthogonal vector component diagrams. The direction is selected at the demagnetization level at which soft components appear to be erased and where the vector direction changes the least. In some cases we also selected the magnetization direction where the vector difference between demagnetization levels agreed with the resultant total vector. The plutonic section is all normally magnetized. This contrasts with the polarity of the overlying dike sequence in which a secondary reversed polarity (in hematite) was detected in approximately one third of the samples [Luyendyk et al., this issue]. The clustering of directions is encouraging. We

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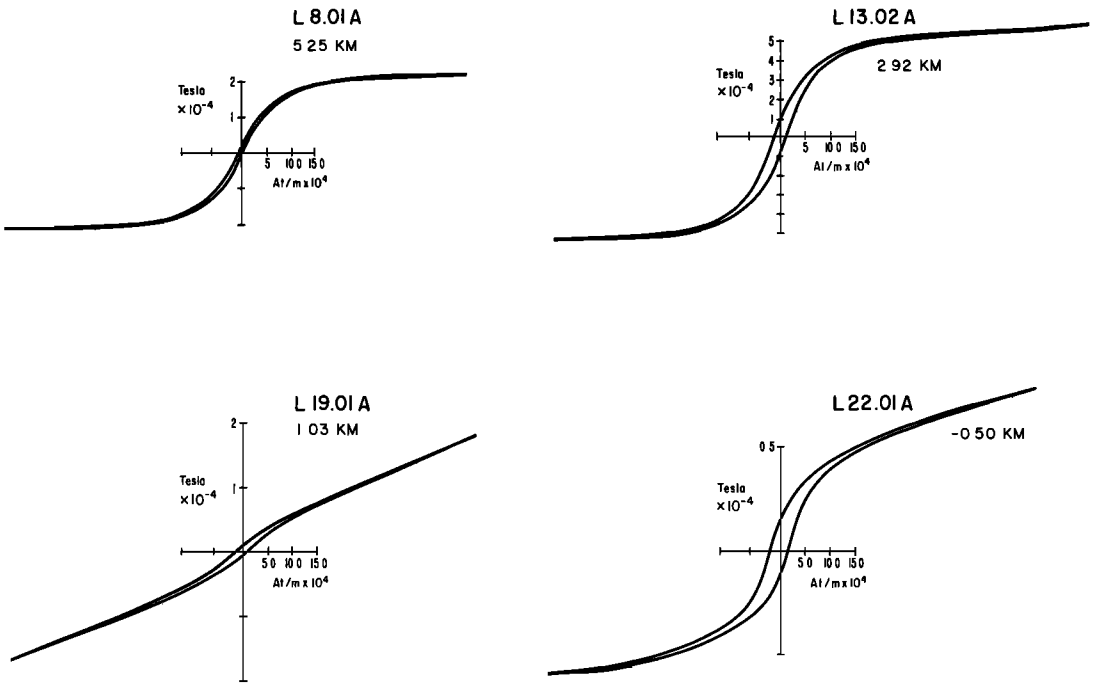


Fig. 6. Representative high-field hysteresis loops for Wadi Kadir samples. L8 is a transitional gabbro, L13 an upper cumulate, L19 a lower cumulate, and L22 a serpentinized harzburgite.

could improve the clustering by selective omission of marginally unreliable samples, such as site L13, which displays some anomalous magnetic properties. Those directions shown in Figure 12 have not been corrected for possible later structural reorganization.

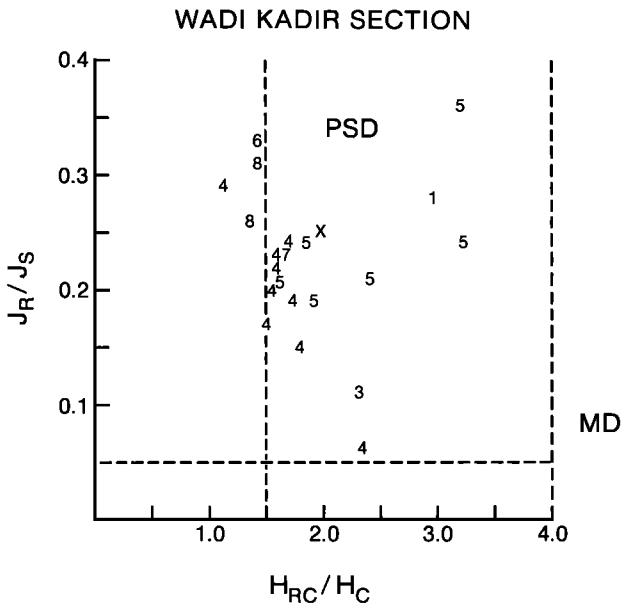


Fig. 7. The ratios J_R/J_S plotted versus H_{RC}/H_C for Wadi Kadir samples. The numerals distinguish rock type: X, dike rock; 1, high-level gabbro; 2, transitional gabbro (none measured); 3, plagiogranite, 4, upper cumulate (above 2 km); 5, lower cumulate; 6, cumulus wehrlite; 7, cumulus dunite; and 8, serpentinized harzburgite.

From field mapping of the ophiolite near Ibra, it is seen that the plutonic section near Wadi Kadir dips about 15°S [Hopson et al., 1981; Pallister and Hopson, 1981]. This structural tilt correction steepens the inclinations of the directions in Figure 12. We averaged directional components from all the gabbro sites together rather than separately. The average structurally uncorrected direction for the gabbros is I = -7.62, D = 339.1, and the structural correction yields I = +6.0, D = 339.0. Directions from site L13 are omitted in the calculations. The uncorrected directions give a paleomagnetic pole at 56.4°N, 81.6°W (n = 45, alpha 95 = 7.55, kappa = 9.12), while the structurally corrected direction shows a pole at 60.5°N, 75°W (Figure 13). The gabbro pole is not different at the 95% confidence level from the poles for the Samail sheeted dikes [Luyendyk et al., this issue]. Relative to the geocentric axial pole, all the Samail poles are far-sided, indicating northward motion of the ophiolite. However, relative to the Cretaceous African pole [Gough et al., 1964], the Samail poles are near-sided. This suggests an interpretation where the ophiolite moved on the order of 10° south to be obducted onto Arabia, then was subsequently carried north with Arabia some 30° or 40°. Of course, this assumes that the Africa poles can be applied to a stable cratonic Arabia.

The stable directions for the Wadi Kadir ultramafic rocks are similar to the stable directions for the gabbros (Figure 12). An uncorrected pole for the ultramafic rocks overlaps the uncorrected gabbro pole at the 95% confidence level (uncorrected ultramafic pole: n = 12, I = -3.03, D = 355.5, 65.2°N, 110.9°W, alpha 95 = 27.7, kappa = 3.41.

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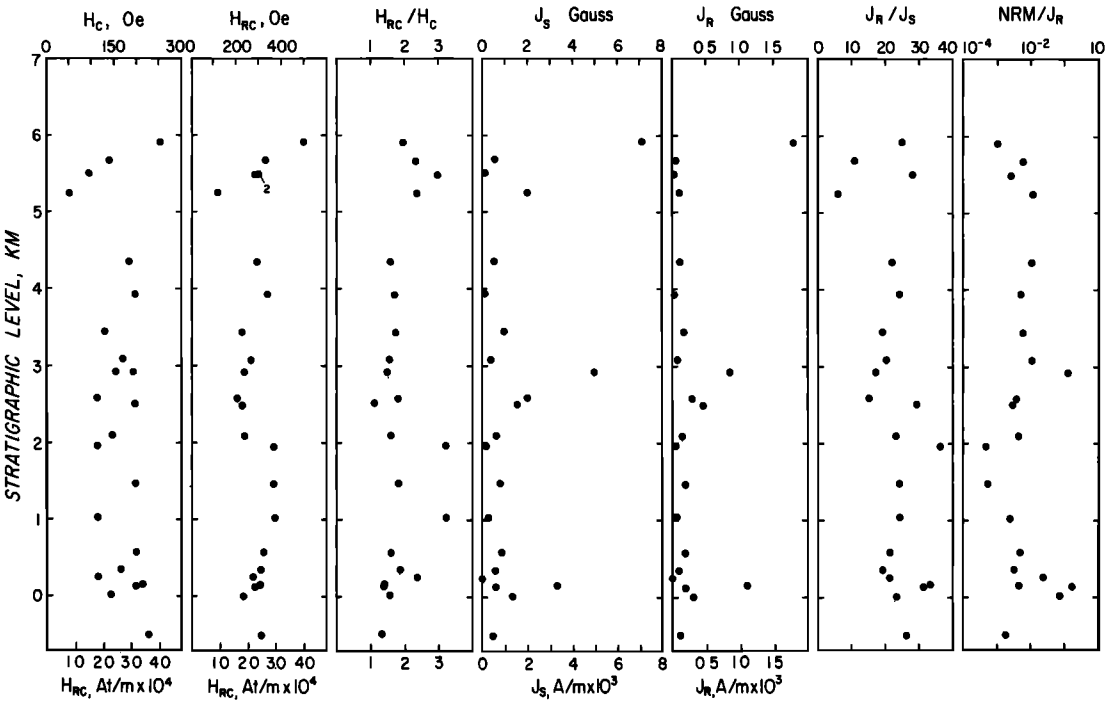


Fig. 8. Rock magnetic properties as a function of stratigraphic level.

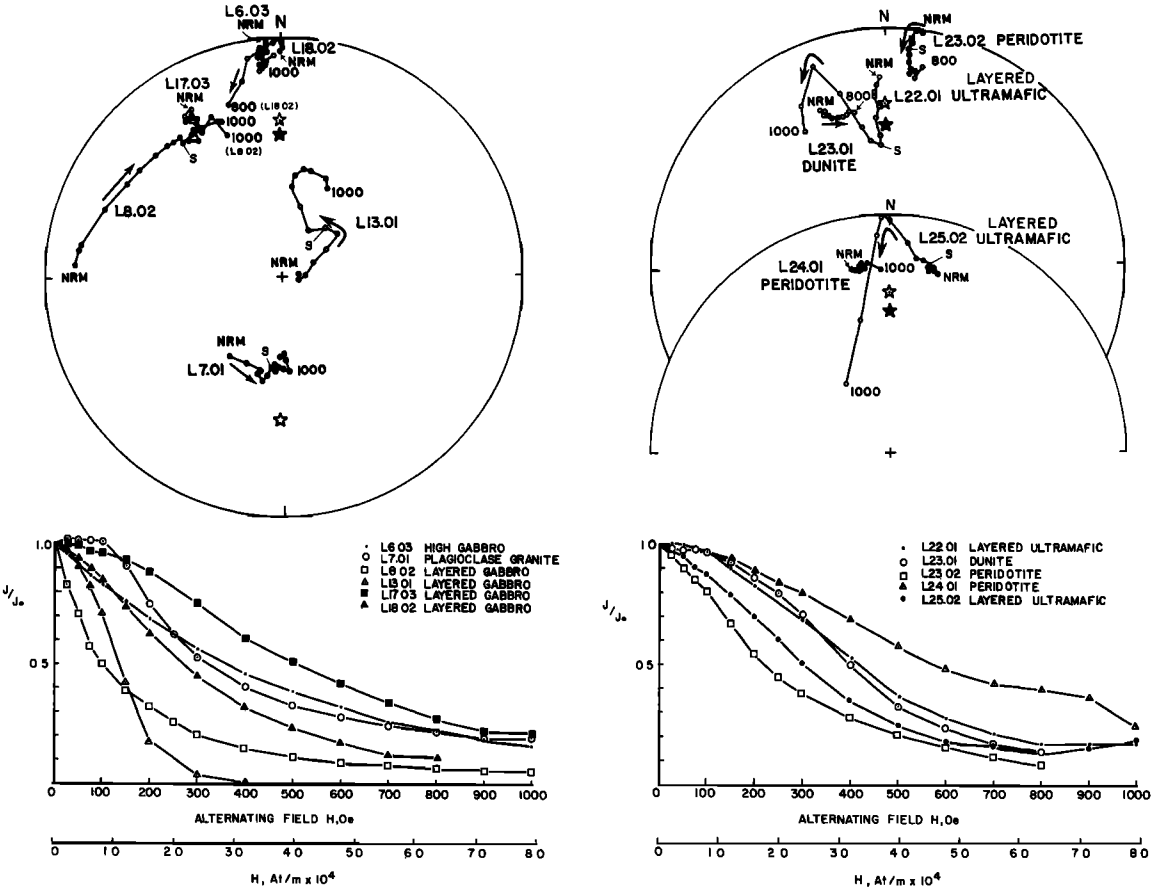


Fig. 9. (left) Representation of AF demagnetization behavior of gabbro samples from Wadi Kadir. On the stereogram, 'S' indicates the stable magnetic direction selected. The solid star and open star are axial dipole directions, while the open star with dot is the present field direction (I = 33, D = 0). (right) Representation of AF demagnetization behavior of ultramafic samples from Wadi Kadir.

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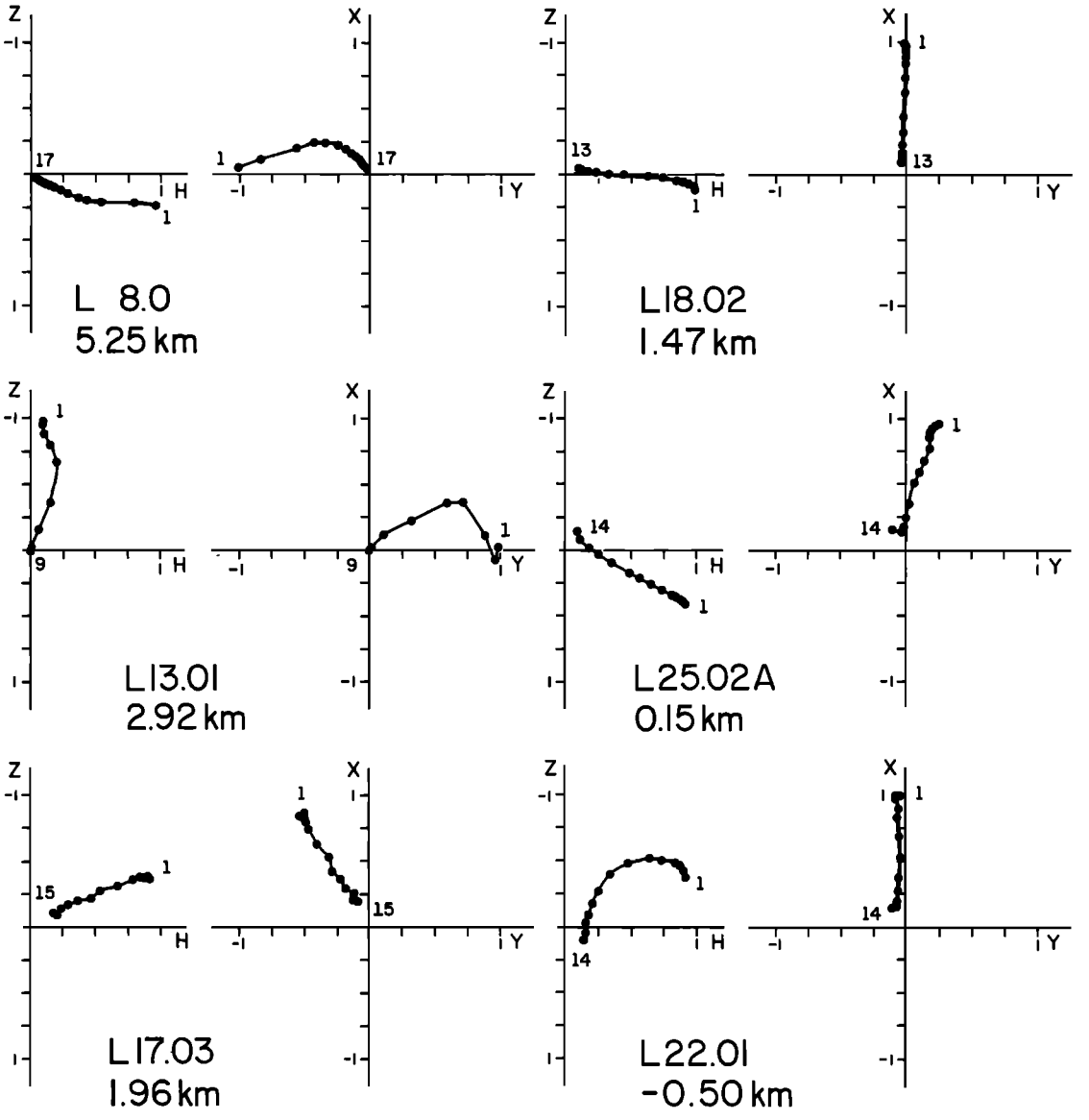


Fig. 10. Orthogonal vector component diagrams for some AF demagnetization series. L8 is a transitional gabbro, L13 an upper cumulate, L17 and L18 lower cumulates, L25 a cumulate wehrlite, and L22 a serpentinized harzburgite.

Discussion and Conclusions

One of our most important observations for the Samail ophiolite is the apparent division of the gabbros into two or more magnetic layers. From relatively precise stratigraphic sampling we find that the NRM, susceptibility, and possibly the magnetic mineral content and grain size decrease in the lowermost 2 km of the cumulate gabbros [Day and Luyendyk, 1979] (Figures 3, 4, and 8). Dunlop and Prévot [1982] and Swift and Johnson [1982] have also proposed or noted magnetic layering in the gabbro sequence, which differs somewhat from what we observe. These authors noted a higher NRM intensity in the upper metagabbros or amphibolite grade gabbros corresponding to the high-level and transitional gabbros of the Samail ophiolite. We observed that the NRM and/or susceptibility of these units at Wadi Kadir are not detectably higher than the cumulate gabbros, although the concentration of opaque oxides in the upper units is higher and these oxides are mostly primary (Figure 4).

The higher concentration of primary oxides in the high-level and transitional gabbros in the Samail ophiolite, and in the ocean crust, is due to the history of crystallization in an oceanic layer 3 magma chamber. Interplay of differentiation, magma replenishment, and mixing result in primary Fe-Ti oxides being found mainly near the roof of the chamber [Pallister and Hopson, 1981]. These factors predict that the thickness of a 'magnetic unit' containing primary oxides would be extremely variable. Subsequent retrograde hydrothermal alteration of the high-level gabbros can also be imagined to have a highly variable effect, with the most extreme being a destruction of magnetic minerals.

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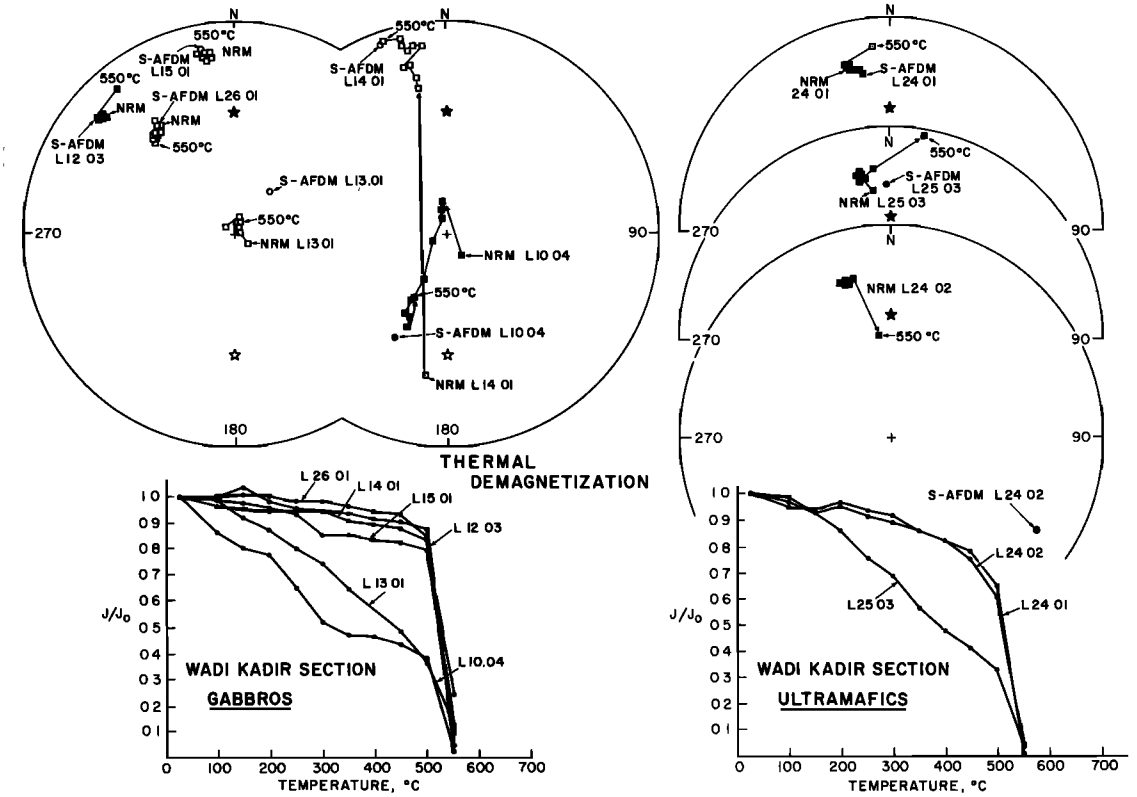


Fig. 11. Representative thermal demagnetization behavior of gabbro (left) and ultramafic (right) samples from Wadi Kadir. On the stereograms the stable direction selected during AF demagnetization of sister samples is shown. The stars indicate the axial dipole field direction.

The magnetic mineral in the Wadi Kadir cumulate gabbros and ultramafic rocks is mainly secondary magnetite from the serpentinization of olivine: this magnetite is virtually absent above 4 km due to a paucity of primary olivine above this level. The serpentinization of olivine is caused by deep penetration of seawater into the ocean crust. Two factors of the magmatic history of the magma chamber serve to control the distribution of secondary magnetite in the cumulate gabbros, cumulate ultramafics, and ultramafic rocks. First, the parent minerals for the paragenesis of serpentine and magnetite, olivine and clinopyroxene, are more abundant in the deeper levels, in the cumulate ultramafics and mantle rocks. Second, chemical analyses show that the concentration of Fe increases with stratigraphic level. For example, the olivines are more forsteritic below 3 km [Pallister and Hopson, 1981, Figures 11 and 15]. Microprobe results from olivine and pyroxenes show that about 24% FeO is found in olivine in the upper levels of the cumulates, compared to 9 to 10% near the base, and that the variation of FeO in pyroxene is from 10% in upper levels of the cumulates to 2 to 6% in the dunites [Pallister and Hopson, 1981, Tables 2 and 4]. Thus these two effects act in an opposite sense: more mafic minerals are available to produce magnetite at the base of the section, but the same minerals contain more iron in the upper section. The result of pervasive hydrothermal circulation in the ocean crust [Gregory and Taylor, 1981] is the production of secondary magnetite (and serpentine), which displays a minimum concentration in the lower levels of the cumulate gabbros (Figure 4) (this may be correlated with the minimum in NRM, and susceptibility at the same level; Figure 3) and a maximum oxide concentration and NRM in the cumulus dunite.

In spite of the higher concentration of primary oxides (exsolved magnetite plus ilmenite) in the high-level gabbros, this rock type exhibits the same or lower NRM intensity as the upper section of cumulates (Figure 3). The significant decrease in NRM intensity, from a crustal volume standpoint, occurs in the cumulates below 2 km due to the drop-off in the concentration of secondary magnetite. Thus the gabbro section is divided into two and possibly three intensity layers by virtue of the magmatic and hydrothermal processes operating within the ocean crust.

Although the secondary magnetite in the Samail gabbros and ultramafics formed at the ridge crest at elevated temperatures, the magnetite in the ultramafics is evidently also due to low-temperature serpentinization by meteoric water [Barnes and O'Neil, 1978]. Therefore it appears that the magnetic properties and directions for the Wadi Kadir ultramafic rocks may not be totally pertinent to a description of the magnetism of the ocean crust and upper mantle.

The stability and intensities of the magnetism of the gabbroic rocks are comparable to the overlying dikes and pillow basalts (Table 2). Also, the coherence of the sample directions is better

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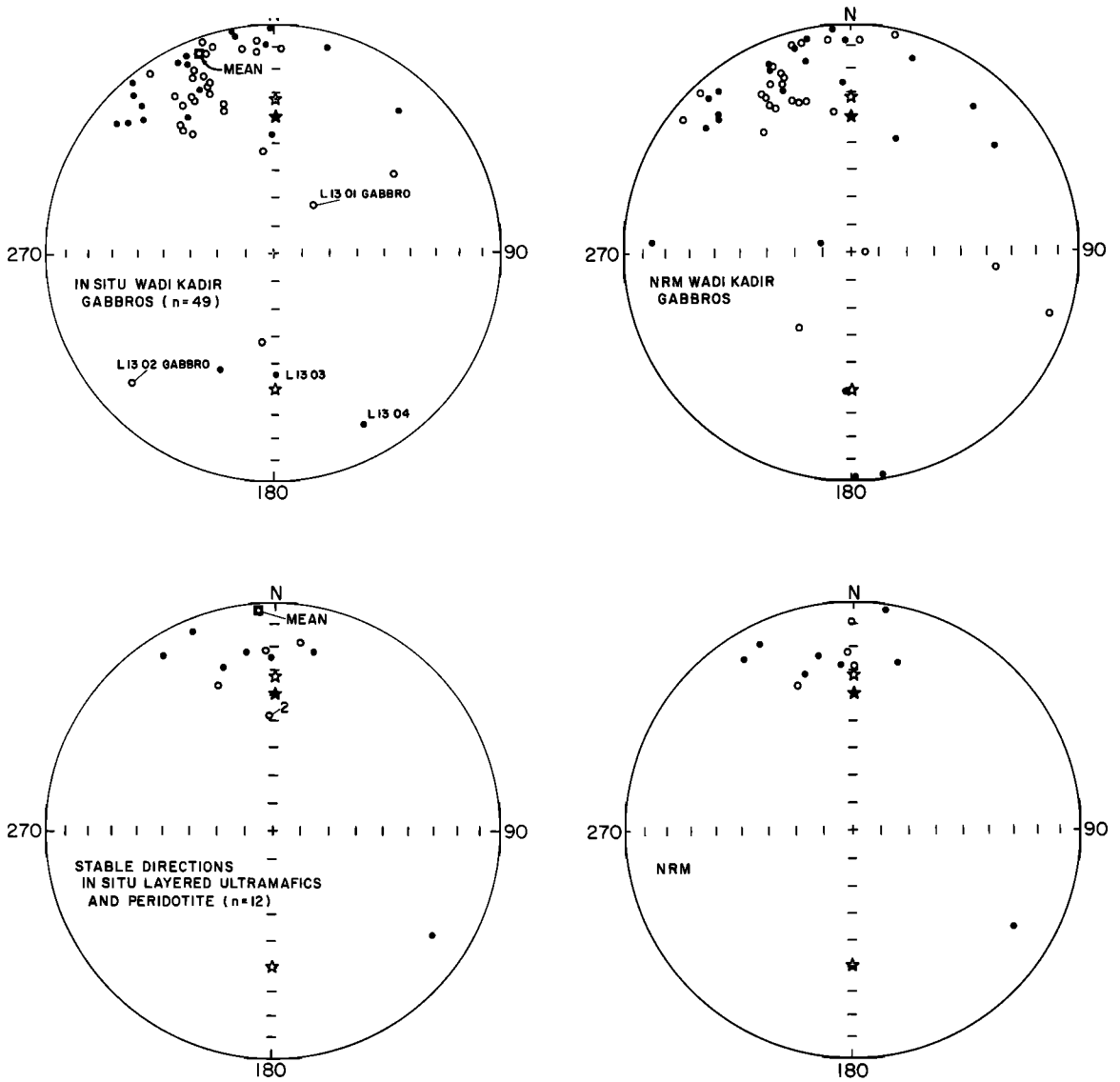


Fig. 12. Stereograms of NRM directions (right) and selected stable directions (left) for gabbro samples (upper) and ultramafic samples (lower) from Wadi Kadir. Directions are in situ and not corrected for a regional attitude of N90°E, 15°N.

than the sheeted dike rocks [Luyendyk et al., this issue]. This coherence of directions shows that the gabbro layer will have on the whole a significant net magnetization.

To evaluate the relative source strengths of the various magnetic layers, we have multiplied the layer NRM intensity by its thickness. This calculation gives a total magnetization in amperes for each meter of cross-strike coordinate. In Table 2 these data are shown for both the Samail and Point Sal ophiolites [Kempner, 1977]. For the Samail ophiolite, the contribution of the upper cumulate gabbro magnetic layer far exceeds that of the pillow basalts or of the combination of the pillows and sheeted dikes. The contribution from the lower cumulate gabbro magnetic layer is not significant, especially considering the added distance of this layer from the sea surface measuring level. For the Point Sal rocks, on the other hand, the extrusive layer total intensity is much greater than that for deeper layers. However, the gabbro section at Point Sal can also be divided into an upper intense layer and a lower weak layer. This relative magnetic 'bottom' we see in the gabbro section of both ophiolites may be a common phenomenon and will have to be considered in the deep layer magnetic modeling schemes of Blakely [1976] and Cande and Kent [1976].

The total vertical component anomaly relief contributed by each layer of thickness Δt can be estimated by $\Delta Z = 2 \cdot 2NRM \cdot \Delta t/(z + \Delta t/2)$ for vertical magnetization where z is the depth to the layer top (Table 2). Using 5 km of water, the upper gabbro section of the Samail ophiolite contributes over half the anomaly amplitude (we omit the calculation for Point Sal rocks because the true thickness of the plutonic section is only approximately known). From Figures 3 and 12 and Table 1 it is evident that ultramafic rocks must be considered as anomaly sources. Careful evaluation of this possibility must consider the

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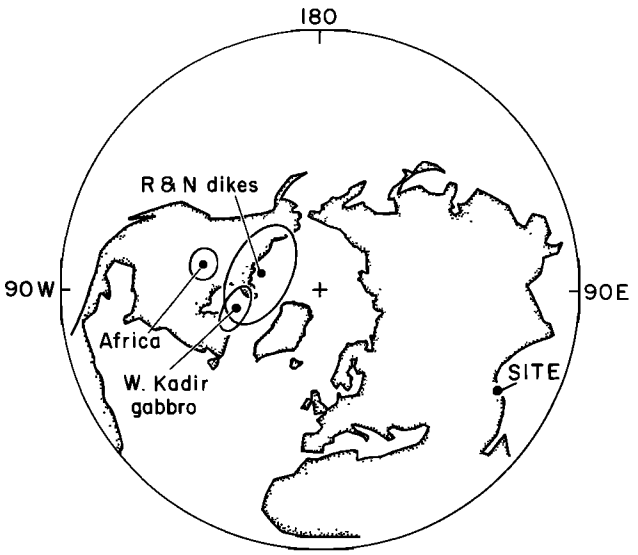


Fig. 13. Comparison of paleomagnetic poles for the Wadi Kadir gabbro samples (corrected for structure), the Samail sheeted dikes [Luyendyk et al., this issue], and the Cretaceous African pole of Gough et al. [1964].

facts that some of the serpentinization (secondary magnetite) may have occurred subaerially, the cumulate ultramafic layer at Wadi Kadir is very thin and would have a volume magnetization of only a few hundred amperes, and the total thickness of the serpentinized harzburgite layer cannot be easily estimated. Thus this possibility remains untested.

A remaining major problem at the present time is to find convincing evidence that the measured properties are indicative of the properties of oceanic layer 3; that is, when was the measured magnetization acquired? Is it a remanence acquired close to the oceanic spreading center? So far, petrographic and isotopic studies of the Samail ophiolite suggest that the magnetism of most of the plutonic section is of chemical origin (CRM due to the production of secondary magnetite at elevated temperature). Banerjee [1980] and Banerjee and Moskowitz [1979] have emphasized the potential importance of an intense stable viscous remanence. In our samples the VRM component appears to be small and is easily removed.

For the Wadi Kadir section of the Samail ophiolite, we can recognize the following important points: (1) the remanent magnetization of the plutonic section is largely a CRM in secondary magnetite, (2) the plutonic section is divided into at least two magnetic layers by serpentinization of minerals with variable Fe content, (3) if the gabbros are magnetized in a restricted time period near the spreading center, then their magnetization intensity and magnetic direction coherence are sufficient to generate about one half the total amplitude of a marine magnetic anomaly, and (4) serpentinized ultramafics are strongly magnetic and can be an anomaly source under special conditions.

The discussions and observations of Dunlop and Prévot [1982] and Swift and Johnson [1982] show that in some circumstances the high-level late-stage gabbros overlying the cumulates may be a more significant magnetic layer than the deeper cumulates. This depends on the magmatic and hydrothermal history of the ocean crust under examination.

Combining the observations from our study and those above, the following points may be fair generalizations on the significance of the plutonic section of the ocean crust in generating marine magnetic anomalies:

1. Magmatic processes lead to the formation of a high-level gabbro containing primary exsolved magnetite near the spreading center. This 'magnetic unit' overlies a cumulate section which may be more weakly magnetized.

2. Hydrothermal circulation (near the spreading center?) permeates the ocean crust to form secondary magnetite in Fe-rich upper portions of the cumulate gabbros, dividing the cumulate gabbro into two magnetic layers in some instances.

3. Hydrothermal circulation may extend into the mantle to form secondary magnetite by serpentinizing peridotite. Thus the mantle must also be considered as a possible anomaly source [Kempner and Luyendyk, 1976; Dunlop and Prévot, 1982].

4. Hydrothermal alteration may be intense enough in the high-level gabbro to destroy their magnetic carriers, leaving the cumulates and ultramafics below as magnetic sources.

5. Given the likely variability of magmatic and hydrothermal processes within the ocean crust from location to location, it seems to us that a generalized magnetic model favoring various portions of the plutonic section as a magnetic source cannot be specified. However, we can identify potential magnetic units as (1) late-stage gabbros with primary Fe-Ti oxides, (2) an upper cumulate gabbro containing secondary magnetite, (3) a lower cumulate gabbro with less secondary magnetite, and (4) ultramafic layers with high concentrations of secondary magnetite.

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